

Joohyun Lee

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EDUCATION

Georgia Institute of Technology, Atlanta, United States
Online Master of Science in Computer Science

Aug 2024 — Present
Overall GPA: 4.00/4.00

- Master's project: 3D registration model with diffusion

Sungkyunkwan University, Suwon-si, South Korea
Bachelor of Mechanical Engineering

Mar 2018 — Feb 2023
Cumulative GPA: 3.85/4.50

WORK EXPERIENCES

Hyundai Motor Company, Seoul, South Korea
Autonomous Driving Division
Research Engineer

Jan 2023 — Present

- **Hyundai RoboTaxi — LiDAR Processing:** Built a CUDA-accelerated LiDAR interface in Python, enabling high-throughput sensor data processing for RoboTaxi perception development and validation.
- **Atria AI — End-to-End Perception:** Designed and developed both camera-based and 3D end-to-end multi-object tracking models in Python, built a custom evaluation pipeline, and improved model performance through architectural redesign and refinement, contributing to the development of Hyundai Motor Group's proprietary autonomous driving model, Atria AI.

Samsung Research, Seoul, South Korea
Robot Center GEMS LAB
University Student Intern

Jul 2022 — Aug 2022

- Action recognition algorithm development based on IMU sensor data mounted on wearable robot
- Outperformed in accuracy with a classification model of boxing moves

LICENSES & CERTIFICATIONS

HDAT (Hyundai Motor Group Data Analytics Test) - Lv.4 (Highest)
Issued by Hyundai NGV

Aug 2023

RESEARCH EXPERIENCES

Human-Augmented Analytics Group (HAAG)

Aug 2025 — Present

Georgia Institute of Technology

Advisors: Dr. Supratik Mukhopadhyay, Dr. Nick Lytle

Project: EvoReg: Versatile and Robust Point Cloud Registration via Multi-Stage Alignment

Developed a coarse-to-fine framework for rigid and non-rigid point cloud registration, with a focus on robustness to noise, partial overlap, and deformation. Contributed throughout the project, including model architecture design, baseline evaluation pipeline development, inference optimization, and manuscript preparation. Demonstrated strong performance across multiple registration benchmarks under a unified evaluation protocol. Submitted to NeurIPS 2026.

Project: LatticeSpike: Real-Time Token-Level Hallucination Detection via Box Embeddings and Spiking Dynamics

Contributed to a real-time hallucination detector that runs alongside a frozen LLM and scores every token as it is generated. Focused on the signal-extraction and box-embedding stages of the pipeline: implemented the causal per-step signal readout from the language model (hidden states, top-k next-token candidates, per-layer logit-lens KL divergence, attention entropy, and effective rank of layer representations). Designed the pipeline visualizations illustrating the signal-readout, box-embedding, and spiking-dynamics stages of the architecture. The resulting detector demonstrated strong performance across multiple hallucination-detection benchmarks. Submitted to ICLR 2027 (under review).

Robotics Innovatory Lab

Aug 2022 — Dec 2022

Sungkyunkwan University

Advisor: Prof. Hyouk Ryeol Choi

Project: Indoor Autonomous Delivery Robot for Capstone Design Built an indoor autonomous delivery robot system using ROS, Python, and a depth camera for perception and navigation. Designed and integrated a console-controller-based tele-operation interface, connecting manual input signals to the robot's control outputs. Performed indoor environment mapping

via Hector SLAM using manual control, and implemented the TEB Local Planner to enable local trajectory planning and navigation in structured indoor spaces.

AMLAB (Automotive Mechatronics Lab)

Mar 2021 — Feb 2022

Sungkyunkwan University

Advisor: Prof. Sung Ho Hwang

Project: 3D LiDAR-Based Multiple Object Tracking for Autonomous Vehicle Perception

Developed a real-time 3D multiple object tracking system for autonomous vehicle perception during a research co-op at AMLAB. Designed and implemented the full LiDAR perception pipeline, including pointcloud preprocessing, RANSAC-based ground removal, voxel downsampling, ROI filtering, DBSCAN clustering, and 3D bounding box extraction. Built a Kalman-filter-based tracker with Hungarian-assignment data association and track lifecycle management, and integrated radar, camera, IMU, and GPS data in ROS 2 for multi-sensor perception. The system was implemented in Python using ROS 2, Open3D, NumPy, and SciPy.

Project: Autonomous Delivery Robot for Hospital Ward Assistance

Developed an autonomous mobile robot system for hospital ward delivery scenarios using LiDAR-based perception, Ackermann-steering path planning, and real-time collision avoidance. Integrated sensor processing and low-level vehicle control on an NVIDIA Jetson Nano platform using Python, ROS, NumPy, and OpenCV. This project was conducted through the Sungkyunkwan University Engineering Talent Development Program and was recognized with the Excellence Award.

RISE LAB (Robotics and Intelligent System Engineering Lab)

Feb 2021 — Nov 2021

Sungkyunkwan University

Advisor: Prof. Hyungpil Moon

Project: Development of a Simulation Drone Competition Utilizing Realistic Environments and Drone Dynamics

Contributed to an ICT Research Center project focused on realistic drone simulation for autonomous flight research and competition. Worked on trajectory generation and optimization based on the minimum-snap algorithm and implemented drone vibration effects induced by motor-output errors to better reflect real flight dynamics in simulation. Developed the system using Python and Unreal Engine. Received a Minister's Award.

SKILLS

- **Programming:** Python, C++
- **Software:** PyTorch, OpenCV, CUDA, Docker, Git, Linux, NumPy, SciPy, ROS

ACTIVITIES

Sungkyunkwan University Hybrid Electric Vehicle Engineers (HEVEN)

Mar 2020 — Feb 2023

Project Leader in 2021, Recognition Team Leader in 2022

- Led software development for the University Student Autonomous Driving Contest, coordinating system integration across perception and vehicle control modules.
- Received the Gold Award (1st Place) in the Technical Idea Category at the University Student Self-Made Car Contest.
- Organized and participated in the Sungkyunkwan University Autonomous Driving Hackathon.
- Led technical study sessions on autonomous driving, covering topics such as perception, planning, and control.

AWARDS & SCHOLARSHIPS

Smart Car Design and Racing Competition - Dean's Award (1st Place)

Nov 2022

Issued by Sungkyunkwan University, College of Information and Communication Engineering

- Team leader
- Design autonomous driving car, implement driving algorithm

Smart Car Design and Racing Competition - Excellence Award (Design)

Jun 2022

Issued by Sungkyunkwan University WE-UP

- Team leader
- Design autonomous driving car, implement driving algorithm

University Student Self-made Car Contest (Technical Idea Category) - Gold Award

Dec 2021

Issued by The Korean Society of Automotive Engineers (KSAE)

- Ensuring visibility in A-pillar blind spots using depth camera
- Implemented eye coordinate detecting algorithm based on Mediapipe face recognition model

- Processed camera image based on detected eye coordinate in order to show a driver proper image from one's perspective

2021 ICT Colloquium - Minister's Award

Nov 2021

Issued by Ministry of Science and ICT

- Development of a simulation drone competition utilizing realistic environments and drone dynamics

Engineering Talent Development Program - Excellence Award

Nov 2021

Issued by Sungkyunkwan University

- Development of a mobile serving robot in Nursing Hospitals

PATENTS

Method and apparatus for evaluating object tracking performance

- Patent No. (KR)1020260011117, Hyundai Motor Company, Jan 20, 2026.
- *Inventors: Joohyun Lee*
- A comprehensive object tracking performance evaluation system for autonomous vehicles that receives tracking data and ground truth annotations, computes multiple evaluation metrics including association score, detection score, and classification score based on correspondence relationships between tracked and ground truth objects, applies differentiated relative weights to these metrics for weighted aggregation, and ultimately derives a composite tracking performance index to enable quantitative assessment of multi-object tracking system performance in complex driving environments.

System for processing data required for object detection based on artificial intelligence and multiple autonomous driving sensors and method implementing the same

- Patent No. (US)19/415,494, (KR)1020250090242, (CN)202511840140.4, Hyundai Motor Company, Dec 10, 2025.
- *Inventors: Joohyun Lee et al.*
- A hierarchical multi-level coordinate transformation system for autonomous vehicle perception that enables scalable, vehicle-agnostic integration of heterogeneous driving sensors (LiDAR, RADAR, cameras) through unified sensor-to-mounting-to-vehicle coordinate conversion with comprehensive preprocessing, postprocessing, and confidence computation.

System of adaptive calibration for autonomous driving and method implementing the same

- Patent No. (US)19/409,994, (KR)1020250042965, (CN)202511849891.2, Hyundai Motor Company, Dec 5, 2025.
- *Inventors: Hyeonjun Lim, Joohyun Lee, Junghyoung Kwon, Minjae Kim, Chansoo Kim*
- A system and method for adaptive calibration in autonomous vehicles using motion and confidence scores to decide online coordinate adjustment.